SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

CAD/CAM & Computer Integrated Manufacturing: CNC G-Codes & FEA

Department: MECHANICAL	**Course / Subject:** CAD/CAM (ME601PC)
Semester: Semester 6	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: 2D/3D Geometric transformations, Wireframe & Solid modeling (CSG, B-Rep), CNC manual part programming (G & M codes), Finite Element Analysis (FEA), and CIM.

Unit 1: Computer Graphics & Hardware

Product lifecycle management (PLM), Raster scan display, 2D and 3D transformations (Translation, Rotation, Scaling, Reflection), Homogeneous coordinates.

Unit 2: Geometric Modeling Techniques

Wireframe modeling, Surface modeling (Bezier, B-Spline, NURBS curves), Solid modeling: Constructive Solid Geometry (CSG) and Boundary Representation (B-Rep).

Unit 3: Numerical Control & CNC Machines

NC vs CNC vs DNC systems, Coordinate systems (Cartesian, Cylindrical), Motion control (Point-to-point, Contouring), Tool compensation (G41/G42), Canned cycles.

Unit 4: CNC Manual Part Programming

G-Codes (G00, G01, G02, G03, G90, G91), M-Codes (M03, M05, M06, M08, M30), Milling and Lathe part program examples, Subroutines, APT language.

Unit 5: FEA & Rapid Prototyping

Finite Element Analysis (FEA) steps: Discretization, Shape functions, Global stiffness matrix assembly, Rapid Prototyping (Stereolithography SLA, Fused Deposition Modeling FDM).